

Supplementary Material

Early parametric gain of a driven anharmonic oscillator: initial-state moments and the leading quantum correction

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Equation, figure and table numbers without the prefix S refer to the main text. Units are $\Omega_0 = 1$ and unit mass throughout. All scripts named below are in the code repository given in the main text.

S1. Numerical methods

S1.1 Quantum branch

The Schrödinger equation $i\hbar_{\text{eff}}\partial_t\psi = [-(\hbar_{\text{eff}}^2/2)\partial_x^2 + V(x,t)]\psi$ is integrated with the second-order Strang splitting $\exp(-iV\Delta t/2\hbar_{\text{eff}})\exp(-i\hbar_{\text{eff}}k^2\Delta t/2)\exp(-iV\Delta t/2\hbar_{\text{eff}})$ on a periodic grid of length $L = 24$ with $M = 4096$ points and $\Delta t = 1.5 \times 10^{-3}$. The kinetic factor is applied in the Fourier representation, where $p = \hbar_{\text{eff}}k$ and $p^2/2\hbar_{\text{eff}} = \hbar_{\text{eff}}k^2/2$. The potential is evaluated at the midpoint $t + \Delta t/2$ of each step, which makes the splitting second order for the time-dependent Hamiltonian; halving Δt changes the results by less than the limits of Table S1. Each recorded state is labeled with the time at the end of its step, in both branches. Moments are recorded every eight steps. The momentum moments are computed in Fourier space, $\langle p^n \rangle = \Sigma_k (\hbar_{\text{eff}}k)^n |c_k|^2 / \Sigma_k |c_k|^2$, where c_k is the discrete Fourier transform of ψ , and the symmetrized covariance $\langle \hat{x}\hat{p} + \hat{p}\hat{x} \rangle / 2 - \langle \hat{x} \rangle \langle \hat{p} \rangle$ is evaluated as $\text{Re} \langle \hat{x}\hat{p} \rangle - \langle \hat{x} \rangle \langle \hat{p} \rangle$, which equals it because $\langle \hat{p}\hat{x} \rangle$ is the complex conjugate of $\langle \hat{x}\hat{p} \rangle$. For the long runs at $A = 0.85$ we use $L = 32$ and $M = 8192$.

S1.2 Observable and rate

The covariance matrix Σ has entries $\text{Var}(x)$, $\text{Var}(p)$ and the symmetrized covariance. Its largest eigenvalue $\sigma_{\text{max}} = (\Sigma_{xx} + \Sigma_{pp})/2 + [(\Sigma_{xx} - \Sigma_{pp})^2/4 + \Sigma_{xp}^2]^{1/2}$ is averaged with a trailing boxcar of 261 recorded points, which spans 3.13 time units, slightly less than the drive period π . The rate λ is one half of the least-squares slope of $\ln \bar{\sigma}_{\text{max}}$ over the fit window. The factor one half converts the growth rate of a second moment into an amplitude rate, so that $\lambda = \mu(A)$ for the harmonic oscillator on long windows. Samples are recorded at $t = 0.012, 0.024, \dots$, so the first point of the trailing average is at $t = 3.132$, one recording interval after the nominal start $t = 3.12$ of every fit window, and the fit uses the points from $t = 3.132$ onward. The moment hierarchy is evaluated at the same recording times and passed through the same average and fit, so that the coefficients of Table II and the measured rates are obtained by one procedure.

S1.3 Fit windows and acceptance criteria

The fixed window is $t \in [3.12, 3.12 + 0.914/A]$. Its length is chosen so that the harmonic covariance grows by the same factor, $\exp(2 \cdot (A/4) \cdot 0.914/A) = e^{0.457} \approx 1.58$, at every A . A rate is accepted if $R^2 \geq 0.98$ (W1) and if the slopes fitted on the two halves of the window differ by at most 10% of their mean (W2). All rates used for quantitative statements pass both criteria. Quantum and classical rates are always fitted on the same window.

S1.4 Classical branch

Trajectories obey $dx/dt = p$, $dp/dt = -\partial_x V$ and are integrated with the classical fourth-order Runge–Kutta method at the step of the quantum branch. The distribution is represented in three ways. Gaussian states use a Gauss–Hermite product rule with 64 nodes per axis. Isotropic states use a product of a radial rule in $u = (x^2 + p^2)/\hbar_{\text{eff}}$ and a uniform angular rule with 96 points. The radial rule is Gauss–Laguerre (48 nodes) for densities proportional to $e^{-u/s}$ and to $(2u - 1)e^{-u}$, Gauss–Legendre on $[0, 2s]$ for the uniform disk, and a single node for a ring. The signed Wigner function of $|1\rangle$ uses the Gauss–Laguerre nodes with weights multiplied by $(2u - 1)$, some of which are negative. The moment-matched distributions are sums of rings with prescribed weights: a point at the origin with weight $1/2$ plus a ring at $u = 2$ with weight $1/2$ has $\langle u \rangle = 1$ and $\langle u^2 \rangle = 2$, as the vacuum Wigner function; two rings at $u = 2$ and 4 with weights $1/2$ have $\langle u \rangle = 3$ and $\langle u^2 \rangle = 10$, as the Wigner function of $|1\rangle$. For long times (Sec. S5) we use Monte Carlo samples whose initial covariance is set to the exact vacuum covariance.

TABLE S1. Convergence checks (Kerr well, $A = 0.2$). Differences are relative to λ_{harm} unless stated.

Check	Result
Grid $L = 24 \rightarrow 32$ and $M = 4096 \rightarrow 8192$, fixed window, Kerr $A = 0.2$, $\hbar_{\text{eff}} = 0.25$ and 0.03	change of $(\lambda_{\text{qm}} - \lambda_{\text{cl}})/\lambda_{\text{harm}} < 10^{-13}$
Step $\Delta t = 1.5 \rightarrow 0.75 \times 10^{-3}$ with recording every $8 \rightarrow 16$ steps (identical recording and smoothing times), every point of Tables S2–S4 and S6	change of $(\lambda_{\text{qm}} - \lambda_{\text{cl}})/\lambda_{\text{harm}}$ between 5×10^{-7} and 8×10^{-7} ; 4/3 of it is quoted as the numerical uncertainty of each point
Gauss–Hermite $64 \rightarrow 96$ nodes per axis, Kerr $A = 0.2$, $\hbar_{\text{eff}} = 0.25$ and 0.03	change of $(\lambda_{\text{qm}} - \lambda_{\text{cl}})/\lambda_{\text{harm}} < 10^{-14}$
Polar (Gauss–Laguerre $\times$ angle) vs Gauss–Hermite, vacuum	agreement of $\lambda_{\text{cl}}/\lambda_{\text{harm}}$ to 10^{-12}
Thermal state, Fock mixture $n \leq 8 \rightarrow n \leq 20$	quantum correction changes by 10% ($\hbar_{\text{eff}} = 0.25$) and 13% (0.125); weight beyond $n = 20$ is 10^{-10}
Hierarchy: finite-difference step $\chi = 10^{-6}$ vs 3×10^{-6}	change of $\kappa_1 < 0.02\%$, of $\kappa_2 < 0.001\%$
Long times, quantum: $M = 4096 \rightarrow 8192$ and Δt halved, $t \leq 250$, $\hbar_{\text{eff}} = 0.06$ and 0.03	change of $\sigma_{\text{max}} < 1.1 \times 10^{-4}$ (relative)
Long times, classical: 64^2 Gauss–Hermite vs 128^2 and Monte Carlo	64^2 departs by $> 1\%$ at $t = 50$ ($\hbar_{\text{eff}} = 0.5$) to 76 (0.03)

S2. Moment hierarchy

In the rescaled variables $X = x/\hbar_{\text{eff}}^{1/2}$, $P = p/\hbar_{\text{eff}}^{1/2}$ the Kerr Hamiltonian is $P^2/2 + \Omega^2(t)X^2/2 + \chi X^4$, and the Wigner–Moyal equation reads $\partial_t W = -P\partial_X W + (\Omega^2 X + 4\chi X^3)\partial_P W - \chi X\partial_P^3 W$. For a centered, inversion-symmetric initial state all odd moments vanish at all times. The exact equations for the second moments are Eq. (4). The fourth moments obey, to the order needed,

$$\frac{d}{dt}\langle X^4 \rangle = 4\langle X^3 P \rangle, \quad \frac{d}{dt}\langle X^3 P \rangle = 3\langle X^2 P^2 \rangle - \Omega^2\langle X^4 \rangle, \quad (\text{S1})$$

$$\frac{d}{dt}\langle X^2 P^2 \rangle = 2\langle X P^3 \rangle - 2\Omega^2\langle X^3 P \rangle, \quad (\text{S2})$$

$$\frac{d}{dt}\langle X P^3 \rangle = \langle P^4 \rangle - 3\Omega^2 \langle X^2 P^2 \rangle + 6\chi \langle X^2 \rangle, \frac{d}{dt}\langle P^4 \rangle = -4\Omega^2 \langle X P^3 \rangle + 24\chi \langle X P \rangle, \quad (\text{S3})$$

where the classical anharmonic contributions to the fourth-moment equations, which involve sixth moments, are of higher order and have been dropped. The terms $6\chi\langle X^2 \rangle$ and $24\chi\langle X P \rangle$ are the Moyal sources. Writing every moment as $M = M^{(0)} + \chi M^{(1)} + \chi^2 M^{(2)} + \dots$ and separating the Moyal-sourced part gives a closed linear system of 19 equations: three for the zeroth-order second moments, five for the zeroth-order fourth moments, three for the first-order second moments (sourced by $-4\langle X^4 \rangle^{(0)}$ and $-8\langle X^3 P \rangle^{(0)}$), five for the Moyal-sourced first-order fourth moments, and three for the quantum part of the second-order second moments. The quantum–classical difference of the covariance to order χ^2 is the last set multiplied by χ^2 . The system is integrated with an eighth-order Runge–Kutta method (relative tolerance 10^{-10}). The rates are extracted from $\sigma_{\max}$ of the covariance $\Sigma^{(0)} + \chi \Sigma^{(1)}$ and $\Sigma^{(0)} + \chi \Sigma^{(1)} + \chi^2 \Sigma^{(2)}_q$ at $\chi = 10^{-6}$ with the same smoothing and fit as in the full calculations, and κ_1 and κ_2 follow by finite differences (script `hierarchy.py`).

For the cosine well, $V = -\Omega^2(t)\cos x$, the expansion of the force to third order gives $-\Omega^2(t)X + 4\chi\Omega^2(t)X^3$ in rescaled variables, and $V''' = -\Omega^2(t)\sin x$ gives the Moyal term $+\chi\Omega^2(t)X\partial_p^3 W$. The hierarchy is the same with χ replaced by $-\chi\Omega^2(t)$ in all first-order source terms. The sextic term of the cosine expansion contributes at order χ^2 to the classical dynamics and equally to the quantum dynamics, so it does not enter κ_2 . Splitting $-\Omega^2(t) = -1 - A\cos 2t$, the static part gives κ_1 contributions equal in magnitude and opposite in sign to the Kerr value (an increase of the early gain), and the modulated part gives a reduction 1.7 times larger at $A = 0.2$.

Independent checks of the hierarchy: (i) for $\chi \rightarrow 0$ the measured classical reduction divided by $\chi\langle u^2 \rangle / \langle u \rangle$ tends to κ_1 for all ten distributions of Fig. 3, and linear extrapolation of the disk and ring families, whose higher-order terms are smallest, gives 1.894–1.907 against $\kappa_1 = 1.895$; (ii) the ratio of the measured quantum correction to $\kappa_2\chi^2$ rises towards 1 as χ decreases (Table S2); (iii) the time-resolved ratio $r(t)$ approaches the hierarchy curve as χ decreases (Fig. 4(a)).

S3. Early-time rates

TABLE S2. Kerr well, $A = 0.2$, vacuum, fixed window. Rates relative to λ_{harm} ; $\Delta = (\lambda_{\text{qm}} - \lambda_{\text{cl}})/\lambda_{\text{harm}}$; ratio = $\Delta/(\kappa_2\chi^2)$, with the numerical uncertainty of each point from Table S1; fit statistics are the worse of the two branches.

$\hbar_{\text{eff}}$	χ	$\lambda_{\text{qm}}/\lambda_{\text{harm}}$	$\lambda_{\text{cl}}/\lambda_{\text{harm}}$	Δ	ratio	R^2	W2 spread
0.5	0.02083	0.8972	0.8656	3.16e-02	$0.921 \pm <0.001$	0.99932	0.035
0.35	0.01458	0.9318	0.9159	1.59e-02	$0.945 \pm <0.001$	0.99972	0.032
0.25	0.01042	0.9530	0.9449	8.11e-03	$0.945 \pm <0.001$	0.99985	0.027
0.18	0.00750	0.9671	0.9630	4.19e-03	$0.942 \pm <0.001$	0.99991	0.023
0.125	0.00521	0.9779	0.9758	2.02e-03	$0.943 \pm <0.001$	0.99994	0.019
0.09	0.00375	0.9844	0.9834	1.05e-03	$0.948 \pm <0.001$	0.99996	0.017
0.06	0.00250	0.9899	0.9894	4.73e-04	0.957 ± 0.002	0.99997	0.015
0.045	0.00187	0.9925	0.9922	2.68e-04	0.963 ± 0.003	0.99997	0.015
0.03	0.00125	0.9951	0.9950	1.20e-04	0.969 ± 0.006	0.99997	0.014

TABLE S3. Kerr well, $A = 0.1$ and 0.4 , vacuum, fixed windows [3.12, 12.26] and [3.12, 5.41]. Points that fail W1 or W2 are not used in the main text.

A	$\hbar_{\text{eff}}$	Δ	ratio	R^2	W2 spread	accepted
0.1	0.5	8.57e-02	$0.745 \pm <0.001$	0.9924	0.324	no
0.1	0.25	2.57e-02	$0.894 \pm <0.001$	0.9995	0.075	yes
0.1	0.125	6.67e-03	$0.926 \pm <0.001$	1.0000	0.012	yes
0.1	0.06	1.57e-03	$0.949 \pm <0.001$	1.0000	0.006	yes
0.4	0.5	1.39e-02	$0.829 \pm <0.001$	0.9983	0.146	no
0.4	0.25	3.74e-03	$0.894 \pm <0.001$	0.9997	0.042	yes
0.4	0.125	9.67e-04	$0.924 \pm <0.001$	0.9998	0.004	yes
0.4	0.06	2.29e-04	0.949 ± 0.004	0.9998	0.012	yes

TABLE S4. Cosine well, $A = 0.2$, vacuum, fixed window. Columns as in Table S2.

$\hbar_{\text{eff}}$	$\lambda_{\text{qm}}/\lambda_{\text{harm}}$	$\lambda_{\text{cl}}/\lambda_{\text{harm}}$	Δ	ratio	R^2	W2 spread
0.5	0.8528	0.8216	3.11e-02	$0.912 \pm <0.001$	0.99753	0.106
0.35	0.9073	0.8874	1.99e-02	$1.191 \pm <0.001$	0.99933	0.018
0.25	0.9461	0.9374	8.63e-03	$1.011 \pm <0.001$	0.99971	0.018
0.18	0.9672	0.9631	4.09e-03	$0.924 \pm <0.001$	0.99986	0.010
0.125	0.9802	0.9783	1.94e-03	$0.909 \pm <0.001$	0.99993	0.002
0.09	0.9870	0.9860	1.02e-03	$0.919 \pm <0.001$	0.99996	0.004
0.06	0.9920	0.9916	4.61e-04	0.938 ± 0.001	0.99997	0.007
0.045	0.9943	0.9940	2.63e-04	0.949 ± 0.003	0.99997	0.008
0.03	0.9963	0.9962	1.18e-04	0.960 ± 0.006	0.99997	0.009

TABLE S5. Classical $\lambda_{\text{cl}}/\lambda_{\text{harm}}$ for the isotropic distributions of Fig. 3 (Kerr well, $A = 0.2$, fixed window), at the values of $\hbar_{\text{eff}}$ in the column heads. $\langle u \rangle$ and $\langle u^2 \rangle$ are the action moments in units of $\hbar_{\text{eff}}$. Values marked with an asterisk fail (W1) or (W2) and are not shown in Fig. 3: Husimi, vacuum at $\hbar_{\text{eff}} = 0.5$ ($R^2 = 0.9913$, spread 0.126); Gaussian, $\langle u \rangle = 3$ at $\hbar_{\text{eff}} = 0.5$ ($R^2 = 0.9459$, spread 0.39).

Distribution	$\langle u \rangle$	$\langle u^2 \rangle$	0.5	0.25	0.125	0.0625	0.03125
Wigner, vacuum	1	2	0.8656	0.9449	0.9758	0.9889	0.9947
Husimi, vacuum	2	8	0.6506*	0.8656	0.9449	0.9758	0.9889
disk, $\langle u \rangle = 1$	1	1.333	0.9347	0.9699	0.9858	0.9931	0.9966
ring, $u = 1$	1	1	0.9557	0.9789	0.9898	0.9950	0.9975
point + ring (1, 2)	1	2	0.9063	0.9557	0.9789	0.9898	0.9950
signed Wigner, $ 1\rangle$	3	10	0.7763	0.9083	0.9595	0.9814	0.9912
Gaussian, $\langle u \rangle = 3$	3	18	0.4083*	0.7653	0.9081	0.9611	0.9826
disk, $\langle u \rangle = 3$	3	12	0.7544	0.8959	0.9528	0.9781	0.9895
ring, $u = 3$	3	9	0.8539	0.9314	0.9675	0.9844	0.9924
two rings (3, 10)	3	10	0.8171	0.9185	0.9623	0.9822	0.9914

TABLE S6. Quantum and Wigner-transport rates for isotropic initial states (Kerr well, $A = 0.2$, fixed window). The thermal state is a Fock mixture truncated at $n \leq 8$ or $n \leq 20$; the classical branch uses the exact Gaussian Wigner function. Ratio = $\Delta/(\kappa_2\chi^2)$ with the vacuum coefficient. All rates pass (W1) and (W2).

State	$\hbar_{\text{eff}}$	$\lambda_{\text{qm}}/\lambda_{\text{harm}}$	$\lambda_{\text{cl}}/\lambda_{\text{harm}}$	Δ	ratio
vacuum	0.25	0.9530	0.9449	8.11e-03	$0.945 \pm <0.001$
vacuum	0.125	0.9779	0.9758	2.02e-03	$0.943 \pm <0.001$
$ 1\rangle$	0.25	0.9164	0.9083	8.15e-03	$0.950 \pm <0.001$
$ 1\rangle$	0.125	0.9615	0.9595	2.01e-03	$0.935 \pm <0.001$
thermal, $\bar{n} = 0.5$, $n \leq 8$	0.25	0.8739	0.8656	8.38e-03	$0.977 \pm <0.001$
thermal, $\bar{n} = 0.5$, $n \leq 8$	0.125	0.9472	0.9449	2.27e-03	$1.058 \pm <0.001$
thermal, $\bar{n} = 0.5$, $n \leq 20$	0.25	0.8732	0.8656	7.64e-03	$0.891 \pm <0.001$
thermal, $\bar{n} = 0.5$, $n \leq 20$	0.125	0.9469	0.9449	2.01e-03	$0.937 \pm <0.001$

S4. Squeezed initial states

For the squeezed vacuum with $\text{Var}(x) = \hbar_{\text{eff}}e^{-2r}/2$ and $\text{Var}(p) = \hbar_{\text{eff}}e^{2r}/2$ the hierarchy is evaluated with the Gaussian moments $\langle X^4 \rangle = 3\langle X^2 \rangle^2$, $\langle P^4 \rangle = 3\langle P^2 \rangle^2$ and $\langle X^2 P^2 \rangle = \langle X^2 \rangle \langle P^2 \rangle$. Table S7 lists the harmonic rate of each state on the fixed window, the quantum and Wigner-transport rates relative to it, the first-order hierarchy value, and the quantum–classical difference with its hierarchy value. The harmonic rate of a squeezed state is about half that of the vacuum on this window, because the covariance ellipse first rotates towards the amplified direction. The first-order term $\kappa_1\chi$ is then not small (0.64 for $r = 0.35$ at $\hbar_{\text{eff}} = 0.25$), and the expansion is not quantitative at these widths. In addition, all four squeezed-state rates fail criterion (W2) on this window (half-window spreads 0.36–0.55, and $R^2 = 0.978$ for $r = 0.35$ at $\hbar_{\text{eff}} = 0.25$), because the alignment makes the growth non-exponential. The entries of Table S7 are therefore effective slopes of $\ln \bar{\sigma}_{\text{max}}$ averaged over the window, not exponential rates, and the table is a qualitative

comparison only. Within it, Wigner transport follows the quantum rate to within 0.8–6% of the harmonic rate of the state.

TABLE S7. Squeezed vacuum states, Kerr well, $A = 0.2$, fixed window, a qualitative comparison since all rates fail W2. $\lambda_{\text{harm}}^{(r)}$ is the harmonic rate of the squeezed state; all rates in the next four columns are relative to it. $\Delta = (\lambda_{\text{qm}} - \lambda_{\text{cl}})/\lambda_{\text{harm}}^{(r)}$.

State	$\hbar_{\text{eff}}$	$\lambda_{\text{harm}}^{(r)}/\lambda_{\text{harm}}$	quantum	Wigner	hierarchy, 1st order	Δ	hierarchy Δ
x-squeezed, $r = 0.35$	0.25	0.549	0.385	0.363	0.355	0.0217	0.0503
x-squeezed, $r = 0.35$	0.125	0.549	0.681	0.673	0.677	0.0077	0.0126
p-squeezed, $r = 0.35$	0.25	0.521	1.284	1.226	1.509	0.0583	0.0697
p-squeezed, $r = 0.35$	0.125	0.521	1.200	1.183	1.254	0.0169	0.0174

S5. Long-time dynamics

Deterministic quadratures of fixed size represent a smooth initial distribution exactly for polynomial moments, but after the ensemble filaments around the period-two states they undersample it. The 64^2 Gauss–Hermite result departs by more than 1% from a 128^2 rule and from Monte Carlo sampling at $t \approx 50$ for $\hbar_{\text{eff}} = 0.5$ and at $t \approx 76$ for 0.03 (Table S1). All long-time classical results therefore use Monte Carlo samples of 4×10^4 trajectories with time step 6×10^{-3} , standardized to the exact initial covariance (script longmc2.py): one sample (seed 11) for $\hbar_{\text{eff}} \geq 0.25$ and the mean of three independent samples (seeds 11, 2 and 3) for $\hbar_{\text{eff}} \leq 0.125$. Standardizing the full initial covariance is necessary. A sample rescaled by a single factor to the exact initial σ_{max} keeps the sampling error of the other second moments, and for 10^5 trajectories its σ_{max} is offset by about 0.5% from the standardized samples and from the 64^2 quadrature already at $t \approx 10$. The sampling uncertainty is measured from the three samples. The relative standard deviation of the period-averaged σ_{max} between samples, up to the first quantum maximum, is at most 0.22%, 0.20% and 0.19% at $\hbar_{\text{eff}} = 0.125, 0.06$ and 0.03 , so the standard error of the mean is at most 0.13%, 0.11% and 0.11%, and 0.05–0.06% at the first maximum itself. Over the whole run the between-sample standard deviation averages 0.3–0.4%. Estimates

from three samples are indicative, and ratios of a deviation to the standard error should not be read as established levels of statistical significance. Before $t \approx 50$ the 64^2 quadrature, which has no sampling error, gives largest quantum–classical differences of 1.11%, 0.45% and 0.18% at $\hbar_{\text{eff}} = 0.125, 0.06$ and 0.03 , close to the Monte Carlo values in Table S8. At $A = 0.2$ and $\hbar_{\text{eff}} = 0.125, 0.06$ and 0.03 the break time from the individual samples differs by at most 0.45, so at these widths the 5% criterion that defines t_b is not affected by sampling. At $A = 0.85$ the classical ensemble is the mean of two standardized samples (script longchaos2.py), and their break times differ by at most 0.15. Where the deviation oscillates close to 5%, t_b is sensitive to the threshold itself: at $A = 0.85$ and $\hbar_{\text{eff}} = 0.25$ a 4% threshold gives 21.7 instead of 25.3. The quantum long runs use the grid of the early-time calculations; for $\hbar_{\text{eff}} = 0.06$ and 0.03 , where σ_{max} reaches its largest values, doubling M and halving Δt changes σ_{max} by less than 1.1×10^{-4} up to $t = 250$. Table S8 lists the break times, the deviation at the first quantum maximum and the largest deviation before it.

Finite-time Lyapunov exponents were computed from the tangent dynamics with renormalization every unit of time, for 80 initial points on circles of radius 0.02–2.4 around the origin (script lyap.py). At $A = 0.2$ the mean exponent is 0.0073 at $T = 500$ and 0.0024 at $T = 2000$ averaged over all points, a decrease close to the $(\ln T)/T$ decay of regular motion. At $A = 0.85$ the points within radius 0.1 of the origin give 0.043 at $T = 1000$, while points at radius 0.8 and beyond give 0.003–0.006, as for regular motion. The break times at $A = 0.85$, from quantum runs with $L = 32$ and $M = 8192$ and the mean of two standardized Monte Carlo samples of 4×10^4 trajectories (seeds 11 and 2, script longchaos2.py), are 19, 25, 28, 33 and 36 at $\hbar_{\text{eff}} = 0.5, 0.25, 0.125, 0.06$ and 0.03 .

TABLE S8. Long-time quantum–classical comparison, Kerr well, vacuum. t_b , first time with a 5% difference of the period-averaged $\sigma_{\max}$; t_{pk} , first classical maximum; dev. at first max., relative difference of the period-averaged $\sigma_{\max}$ at the first quantum maximum; largest dev. before it, the largest relative difference between the first smoothed point and the first quantum maximum, with its time in parentheses. Classical reference, the mean of the standardized Monte Carlo samples of Sec. S5.

$\hbar_{\text{eff}}$	t_b	t_{pk}	dev. at first max.	largest dev. before it (t)	first quantum min., t	$\sigma_{\max}/\hbar_{\text{eff}}$ there
0.5	25.7	31.4	1.1%	4.63% (16)	40.8	0.56
0.25	40.8	38.6	8.6%	8.64% (44)	77.2	0.60
0.125	69.7	46.3	0.30%	1.14% (20)	69.5	4.41
0.06	84.6	53.2	0.11%	0.49% (24)	99.0	3.84
0.03	96.5	59.8	0.13%	0.21% (29)	138.2	2.45

S6. Cosine well: escape and barrier height

For a potential with a single length scale and fixed drive amplitude, the barrier height in quanta, $N_b = V_b/(\hbar_{\text{eff}}\Omega_0)$, and χ are not independent. Rescaling gives $N_b\chi = 1/16$ for the softening quartic well $\frac{1}{2}\Omega_0^2x^2 - (\beta/4)x^4$ and $1/12$ for the cosine well. This is a scaling property of each potential. Adding a sextic term $(\gamma/6)x^6$ to the softening quartic well changes V_b at fixed $|U|$. For $\gamma < 0$ the barrier position $x_b^2 = u^* = [(\beta^2 + 4|\gamma|\Omega_0^2)^{1/2} - \beta]/(2|\gamma|)$ moves continuously inward and V_b decreases towards zero. For $0 < \gamma \leq \beta^2/(4\Omega_0^2)$ the barrier rises, up to a factor $4/3$ at $\gamma = \beta^2/(4\Omega_0^2)$, where the maximum and the outer minimum merge into an inflection point. The sextic term therefore tunes N_b at fixed χ over $(0, 4/(3 \cdot 16\chi)]$.

Escape from the cosine well is studied with classical transport of the vacuum Wigner ensemble (polar quadrature, 32×96 points), with escape defined as reaching $|x| > \pi$. Figure S1(a) shows the escaped fraction by $t = 60$. For $\hbar_{\text{eff}} \leq 0.25$ it stays below 6% up to $A = 0.55$ and rises to about one half at $A \approx 0.65$. The threshold depends on the observation time (Fig. S1(b)). At $\hbar_{\text{eff}} = 0.125$ half of the ensemble has escaped at $A \approx 0.66$ by $t = 60$, at $A \approx 0.57$ by $t = 200$ and at $A \approx 0.53$ by $t = 400$. Escape in the driven cosine well is the escape of the parametrically excited pendulum, whose global bifurcations and chaotic transients are described in Refs. [29,30] of the main text. It is a classical process, and the initial width affects it mainly through the spill-over of the initial distribution at large $\hbar_{\text{eff}}$.

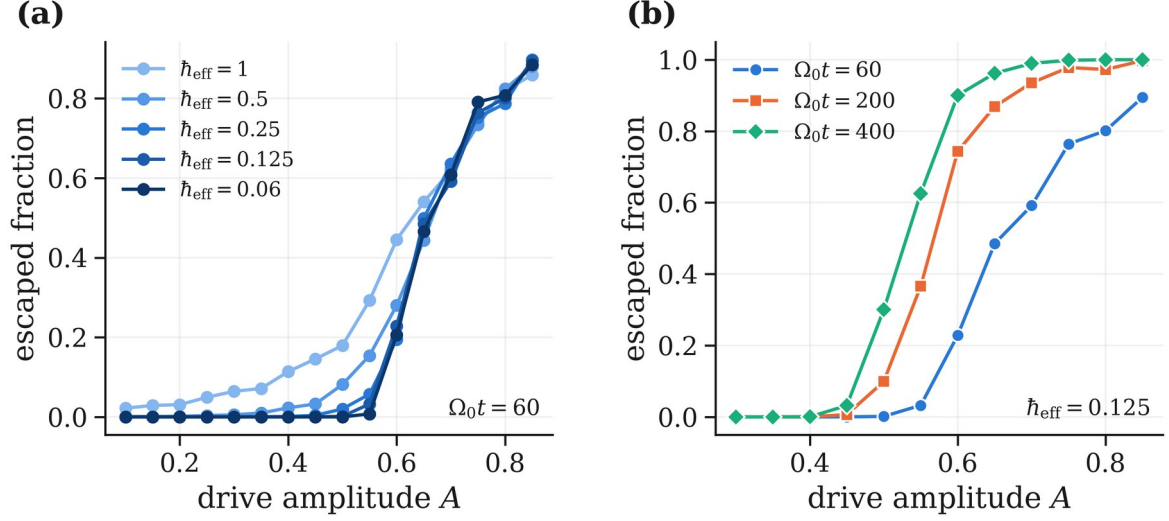


FIG. S1. Classical escape from the driven cosine well, vacuum-Wigner ensemble. (a) Fraction reaching $|x| > \pi$ by $t = 60$ versus A for $\hbar_{\text{eff}} = 1, 0.5, 0.25, 0.125$ and 0.06 (light to dark). (b) Fraction at $\hbar_{\text{eff}} = 0.125$ by $t = 60, 200$ and 400 .

S7. Code

The script `run_all.sh` runs every calculation in order from an empty results folder and then builds all tables and figures; `README.md` lists inputs, outputs and running times. Main scripts: `resid.py` (grid and quadrature engines, fit), `hierarchy.py` and `hier_cos.py` (moment hierarchy), `p01_early.py` (fixed-window rates for both wells and three drive amplitudes), `p02_conv.py` (convergence), `p03_states.py` and `collapse.py` (initial distributions), `cosq.py` (cosine well), `rwa_check.py` (rotating-wave comparison), `longtime.py`, `longmc.py`, `longmc2.py`, `longchaos.py` and `longtime_vals.py` (long times), `lyap.py` (finite-time Lyapunov exponents), `escape_map.py` and `escape_time.py` (escape), `floquet.py` (exact Floquet exponent), `ms_numbers2.py` and `sm_tables.py` (numbers and tables), and `figs/fig1.py`–`fig4.py` and `figS1.py`.